



Loops That Make Patterns

Quick facilitation notes & answer key

Goal

Students see that a repeat loop runs every block inside it each pass, so a loop with two or more actions builds a repeating pattern. They count, change, and write pattern loops.

Curriculum (Ontario Math 2020, Strand C3)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

Materials

A block-coding app (Scratch-style) to project; the printed worksheet + a pencil per student; optional coloured pencils for the pattern.

Run it (about 30 min)

1. Hook: show a bead bracelet or AB pattern — name the part that repeats.
2. Project the repeat-3 (red, blue) loop; stamp it out together, counting beads.
3. Students do Exercises 1–4 (Ex 4 is the stretch), solo or in pairs.
4. Share answers; land the big idea: the loop repeats the WHOLE inside each pass.

Answer key (checked by the run-gate)

Ex 1 — 2 beads each repeat; 6 beads total (2×3).

Ex 2 — 8 beads (2×4); predicted = 8.

Ex 3 — repeat 5 (makes 10 stamps).

Ex 4 (challenge) — 9 stamps (3×3); pattern ●▲■ ●▲■ ●▲■.

Watch for & differentiate

Common slip: counting only one stamp per repeat instead of all the inside blocks.

Support: tap each inside block as you say the repeat out loud.

Extend: ask how many beads if repeat 4 had THREE stamps inside (12).

Success looks like

The student says how many stamps run per repeat and in total (Ex 1–2), writes a repeat number to make a pattern of a set length (Ex 3), and reasons that all the inside blocks repeat together (Ex 4).